Foundations of Computer Science (26CSE0103)

Topic: Software and Programming Language

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Introduction to Software

Software is a collection of programs, procedures, and instructions that enable a computer to perform specific tasks. Unlike hardware, software is intangible and controls how hardware operates.

Types of Software

- System Software
- Application Software
- Utility Software
- Middleware
- Firmware

Key Points

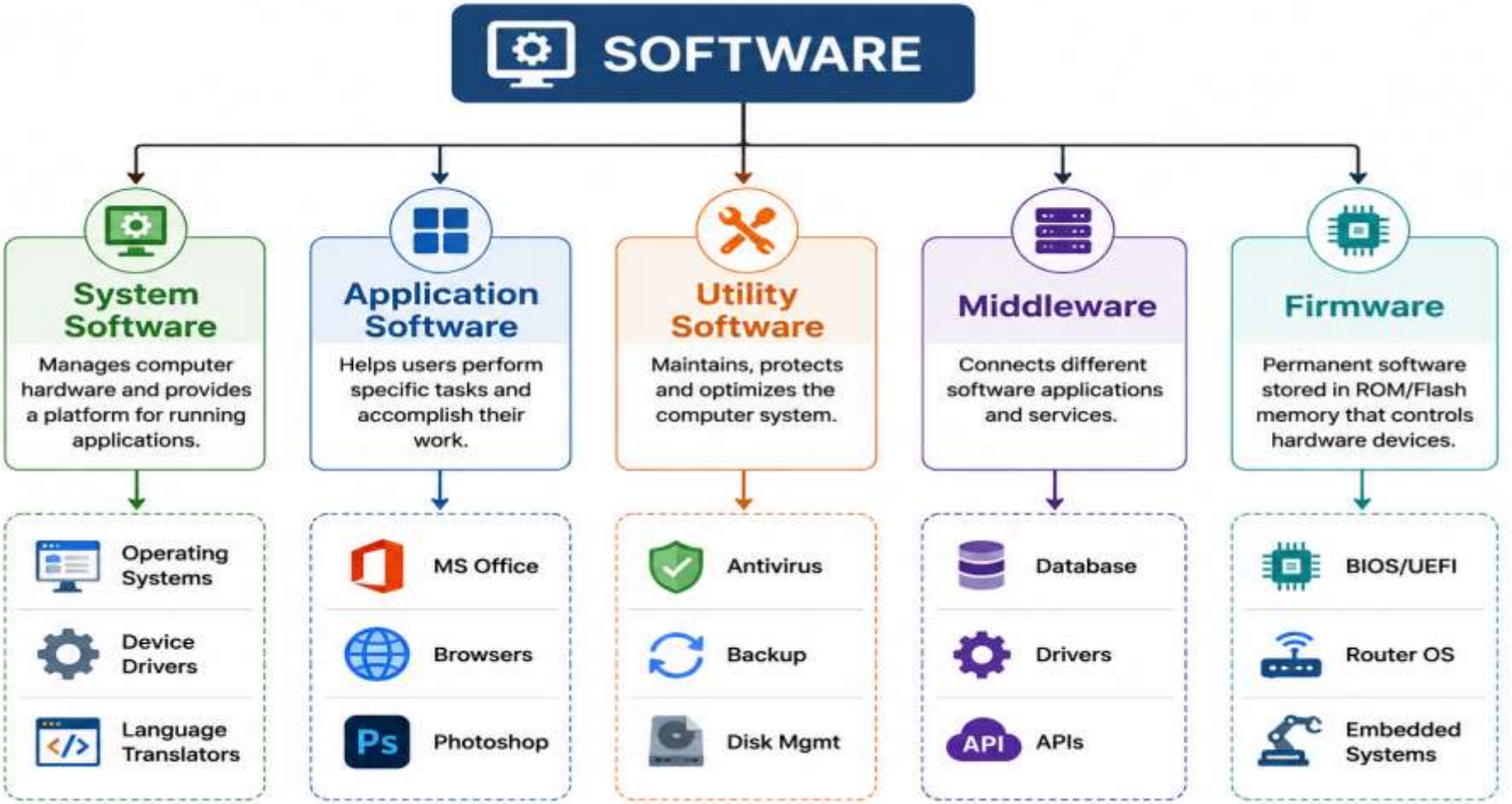
Software controls hardware.

It helps users interact with computers.

Different software serves different purposes.

Software Classification

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System Software

Definition

- System software is software that manages computer hardware and provides a platform for application software.

Functions

- Manages hardware resources
- Controls CPU and memory
- File management
- Device management
- Process scheduling
- Security management

Examples

- Windows
- Linux
- macOS
- Android
- UNIX

Components of System Software

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Operating System

- Manages all hardware resources.

Device Drivers

- Enable communication between operating system and hardware devices.
- Examples:
- Printer Driver
- Graphics Driver
- Network Driver

Language Translators

- Compiler
- Interpreter
- Assembler

Advantages and Disadvantages of System Software

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Advantages

- Efficient hardware utilization
- Resource management
- Security
- User interface
- Multitasking

Disadvantages

- Expensive development
- Hardware dependent
- Complex maintenance
- Vulnerable to malware

Application Software

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Definition

- Application software is designed to help users perform specific tasks.

Categories

- General Purpose
- MS Word
- Excel
- PowerPoint
- Chrome
- Customized Software
- Banking Software
- Hospital Management
- ERP Systems

Types of Application Software

Productivity Software

- Word Processor
- Spreadsheet
- Presentation

Business Software

- ERP
- CRM
- Accounting

Multimedia Software

- VLC
- Photoshop
- Premiere Pro

Educational Software

- Moodle
- Google Classroom

Entertainment Software

- Games

Advantages and Disadvantages of Application Software

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Advantages

- Increases productivity
- User friendly
- Automation
- Saves time
- Accuracy

Disadvantages

- Costly
- Requires updates
- Compatibility issues
- Security risks

Utility Software

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Definition

- Utility software helps maintain, optimize, secure, and improve computer performance.

Functions

- Disk cleanup
- Antivirus protection
- File compression
- Backup
- Recovery

Types of Utility Software

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Antivirus

- Windows Defender
- Norton
- McAfee

Backup Software

- Acronis
- Google Backup
- **Compression**
- WinRAR
- 7-Zip

Disk Utilities

- Disk Cleanup
- Disk Defragmenter

Firewall

- Windows Firewall

Advantages and Disadvantages of Utility Software

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Advantages

- Improves performance
- Protects from malware
- Optimizes storage
- Data recovery
- System maintenance

Disadvantages

- Uses system resources
- Some utilities are paid
- Incorrect use may delete files
- Frequent updates required

Middleware

Definition

- Middleware is software that acts as a bridge between operating systems, applications, databases,and networks.



Types of Middleware

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Database Middleware

- JDBC
- ODBC

Message Middleware

- RabbitMQ
- Apache Kafka

Web Middleware

- Apache Tomcat
- Nginx

Cloud Middleware

- AWS Middleware
- Azure Middleware

Enterprise Middleware

- IBM WebSphere
- Oracle Fusion Middleware

Advantages and Disadvantages of Middleware

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Advantages

- Easy communication
- Scalability
- Integration
- Security
- Flexibility

Disadvantages

- Cost
- Complex implementation
- Maintenance
- Performance overhead

Firmware

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Definition

- Firmware is permanent software stored in ROM, EEPROM, or Flash memory that controls hardware devices.

Characteristics

- Embedded
- Low-level programming
- Hardware specific
- Rarely changed
- Examples
- BIOS
- UEFI
- Router Firmware
- Smart TV Firmware

Firmware Working and Application

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Applications

- Washing Machines
- Printers
- Routers
- Cars
- Smartphones
- Cameras



Comparison Table

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Feature	System Software	Application Software	Utility Software	Middleware	Firmware
Purpose	Manage Hardware	User Tasks	System Maintenance	Connect Applications	Control Hardware
Runs On	Computer	Operating System	Operating System	Server	ROM/Flash
User Interaction	Low	High	Medium	Low	None
Examples	Windows	MS Word	Antivirus	Apache Tomcat	BIOS

Real-Life Examples

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Laptop

- Windows → System Software
- MS Word → Application Software
- Windows Defender → Utility Software
- JDBC → Middleware
- BIOS → Firmware

Smartphone

- Android → System Software
- WhatsApp → Application Software
- Cleaner App → Utility Software
- Google Play Services → Middleware
- Bootloader → Firmware

Sample Questions (MCQS)

Software like WinZip, 7-Zip, and disk cleanup tools fall under:

A) Application Software

B) Utility Software

C) System Software (OS)

D) Firmware

Question MCQS based

A program written in machine language consists of:

A) English-like statements

B) Hexadecimal mnemonics

C) Binary digits (0's and 1's)

D) Standard SQL queries

Scenario based MCQS

Scenario: A hospital updates its MRI machine software. The manufacturer sends a flash tool to update the code stored on the machine's internal control chip to improve sensor calibration.
What level of software was updated?

A) Utility Software

B) Application Software

C) Firmware

D) Middleware

Solutions

1. Utility software
2. Binary digits (0’s and 1’s)
3. firmware

- Which type of software manages computer hardware and provides a platform for application software?
  A. Application software
  B. System software
  C. Middleware
  D. Programming language
- Which of the following is an example of application software?
  A. Operating system
  B. Device driver
  C. Microsoft Word
  D. Compiler
- What is the primary purpose of middleware?
  A. To edit documents
  B. To connect and facilitate communication between different software components
  C. To manage computer hardware directly
  D. To create graphics
- Which of the following is a system software?
  A. PowerPoint
  B. Photoshop
  C. Windows operating system
  D. Web browser
- A compiler is mainly used to:
  A. Manage files
  B. Translate high-level programming language into machine code
  C. Browse the internet
  D. Protect the computer from viruses

- Which category does Python belong to?
  A. System software
  B. Application software
  C. Programming language
  D. Middleware
- Which of the following best describes application software?
  A. Software that controls hardware resources
  B. Software designed to perform specific tasks for users
  C. Software that translates source code only
  D. Software that connects two networks
- Which software acts as an intermediary between an operating system/database and applications?
  A. Middleware
  B. Word processor
  C. Compiler
  D. Device driver
- Which of the following is **not** a programming language?
  A. Java
  B. C++
  C. Python
  D. Linux
- Which statement correctly differentiates system software from application software?
  A. System software is only used for entertainment.
  B. Application software manages hardware directly.
  C. System software supports computer operation, while application software performs user-oriented tasks.
  D. Both are exactly the same.

Answer Key

Q	Answer
1	B
2	C
3	B
4	C
5	B
6	C
7	B
8	A
9	D
10	C

1. Scenario: A college develops a Java-based student management system. The application needs to connect to a MySQL database, execute SQL queries, retrieve student records, and update marks. The developer wants a Java-specific database connectivity mechanism.
Which technology should be used?
A. ODBC
B. JDBC
C. Middleware only
D. System software

2. Scenario: A company has developed an application that needs to communicate with databases from different vendors. The developers want a **standard interface** so that the application can access different databases by using the appropriate database-specific drivers, without being tightly dependent on one database vendor.
Which technology best fits this requirement?
A. JDBC
B. ODBC
C. Operating System
D. Compiler

3. Scenario: A banking application communicates with a database, authentication service, and payment service. Instead of allowing the application to communicate directly with every service, an intermediate software layer manages communication, data transformation, and coordination between these different components.
What type of software is primarily being used?
A. Application software
B. System software
C. Middleware
D. Programming language

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Thank You